BT5240 - Computational Systems Biology
Assignment 4

Due date: April 28, 2026, 17:00 hrs
(Project track will get relaxation till May 1, 2026)

Instructions

If you need any assistance with computing, feel free to approach the TAs. Evaluation will be based on the code(s), the answers, and the methodology.
Academic Integrity: Discussing problems verbally with friends is allowed, but copying or looking at codes is not permitted (use of ChatGPT/LLMs is also NOT permitted). Transgressions are easy to find and will be reported and dealt with strictly. Mention any collaboration (discussions only!) in your solutions. Late submission penalties: 1 second to 24 h: 20%; 24 to 48 h: 40%; >48h: 60%. Early submission bonuses: >24h: 10%, >48h: 20%

Submission

Join the class on Turnitin using class code **52127280** and enrollment key *bt5240_2026*. You need to make two submissions: a Python file and a written report.

1. Code guidelines:
    a. Submit a single *.py* file containing the code for all the questions.
    b. Ensure the code is clean and properly formatted.
    c. Use clear comments to explain logic when necessary.

2. Report guidelines:
    a. Prepare a report that includes answers to all the questions, all required plots/tables properly labelled, clear explanation of the results. Submit a single PDF file.
    b. Use Times New Roman, font size 12, 1.5 line spacing, with clean and consistent formatting.

Total marks: 20

Question 1 (10 marks):
Consider a bioreactor system in which glucose is the feed and lactate is the desired product. Glucose can also undergo a different pathway that produces undesirable byproducts. The biochemical pathway can be simplified as:

R1: Glucose (G) → Pyruvate (Py)
R2: Pyruvate (Py) → Lactate (L)
R3: Glucose (G) → Undesired products (W)

Reactions R1 and R2 follow first-order kinetics, while R3 follows second-order kinetics. The corresponding rate constants are k_1, k_2, and k_3.

(a) Write the rate expressions for all three reactions and the system of ordinary differential equations (ODEs) governing the concentrations of glucose, pyruvate, lactate, and waste products.

(b) Simulate this system from t=0 to t=10 minutes using `solve_ivp`, and plot the concentration profiles of all species. Use the following parameters and initial conditions:

- k_1 =1.0 min^{-1}
- k_2 = 0.5 min^{-1}
- k_3 = 0.2 $Lmol^{-1}min^{-1}$
- Initial concentrations:
  - Glucose = 1.0 mol/L
  - Pyruvate = Lactate = Waste products = 0

(c) Comment on the observed concentration profiles.

(d) Now assume that reaction R1 follows Michaelis-Menten kinetics with K_m=0.5 mol/L and v_{max}=1.2 mol $L^{-1}min^{-1}$. Write the modified rate expression and the corresponding ODE system. Simulate this system over the same time interval and compare the output with the first-order model.

(e) Analyze how the system changes for K_m values: 0.2, 1.0 and 2.0?

Question 2 (10 marks):

Consider a simple Boolean network capturing cell cycle progression in mammalian cells. The network consists of four components: Cyclin D (CycD), Retinoblastoma protein (Rb), E2F transcription factor (E2F) and Cyclin E (CycE). The interactions are described as follows:

- CycD is activated by itself and is inhibited by CycE
- Rb is active when CycD is absent
- E2F becomes active only when Rb is inactive
- CycE is activated by E2F

Each component can be in an active (1) or inactive (0) state. Assume synchronous updates.

(a) Capture the given interactions in a network diagram.

(b) Write the Boolean update rules for all variables.

(c) Construct the state transition table for all possible states.

(d) Simulate the system from all possible initial states and identify attractors.

(e) Plot the evolution of each variable for all the simulations and comment on your observations.